

Common Mathematical Notation

Superscripts

$$2x^3$$

$$2x^34$$

$$2x^{34}$$

$$2x^{3x+4}$$

$$2x^{3x^4+5}$$

Subscripts

$$x_1$$

$$x_12$$

$$x_{12}$$

$$x_{1_2}$$

$$x_{1_2_3}$$

$$a_0, a_1, \dots, a_{100}$$

Greek letters

$$\pi$$

$$\Pi$$

$$\alpha$$

$$A = \pi r^2$$

Trig functions

$$y = \sin x$$

$$y = \cos x$$

$$y = \csc \theta$$

$$y = \sin^{-1} x$$

$$y = \arcsin x$$

Log functions

$$y = \log x$$

$$y = \log_5 x$$

$$y = \ln x$$

Roots

$$\sqrt{2}$$

$$\sqrt[3]{2}$$

$$\sqrt{x^2 + y^2}$$

$$\sqrt{1 + \sqrt{x}}$$

Fractions

$$\frac{2}{3}$$

about $\frac{2}{3}$ of the glass is full.

(you can make the fraction big like that : $\frac{2}{3}$)

Or it easy to type it like this : $\frac{2}{3}$ but you have to `\usepackage{amsmath}`.

$$\frac{\sqrt{x+1}}{\sqrt{x+2}}$$

$$\frac{1}{1 + \frac{1}{x}}$$